

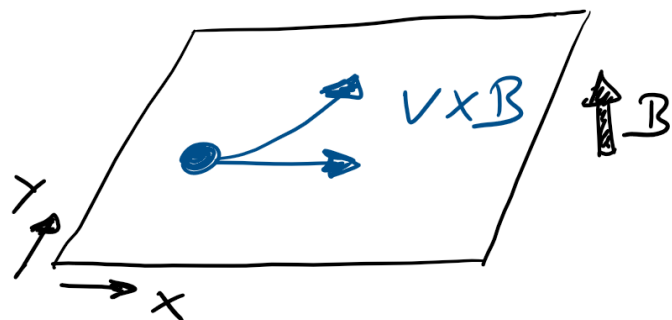


Quanten-Hall-Effekt

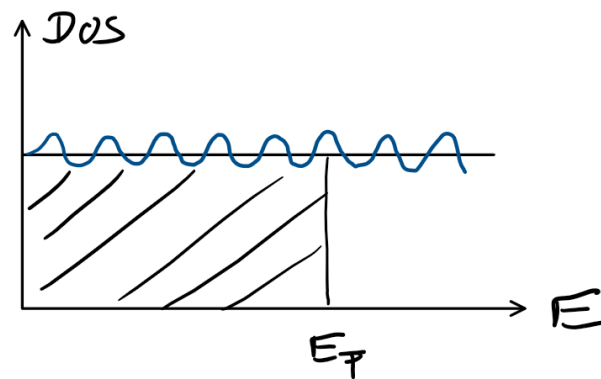
Dr. Christoph Kastl
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Zentrum für Nanotechnologie
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2D Elektronen im Magnetfeld

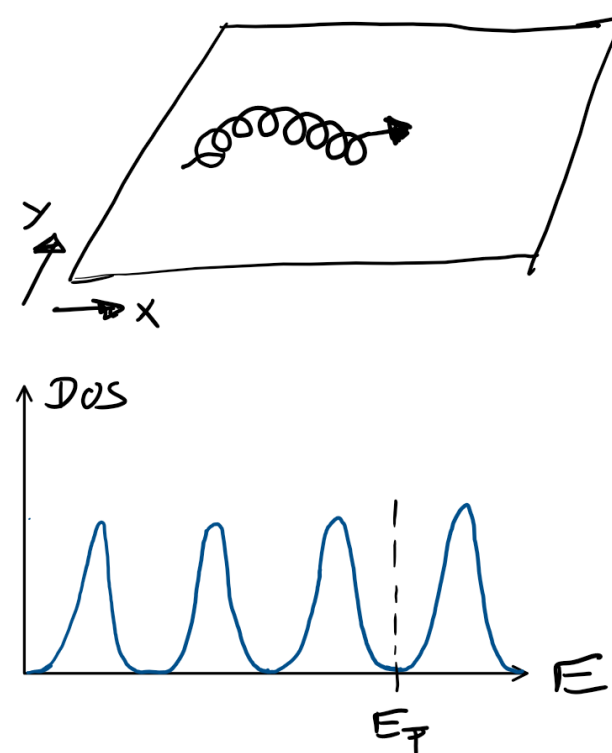
Hall-Effekt (klassisch)



SdH-Oszillationen (semi-klassisch)



Quanten-Hall-Effekt (voll quantisiert)



Klaus von Klitzing

Facts



Photo from the Nobel Foundation archive.

Klaus von Klitzing
Nobel Prize in Physics 1985

Born: 28 June 1943, Schroda, German-occupied Poland (now Poland)

Affiliation at the time of the award: Max-Planck-Institut für Festkörperforschung, Stuttgart, Germany

Prize motivation: “for the discovery of the quantized Hall effect”

2D Halbleiter mit hoher Reinheit und Mobilität

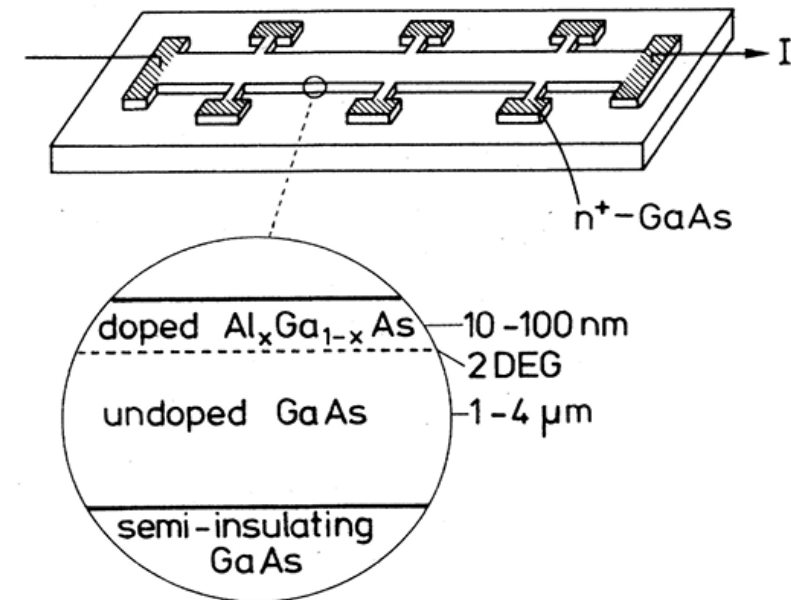


FIG. 3. Typical shape and cross section of a GaAs- $\text{Al}_x\text{Ga}_{1-x}\text{As}$ heterostructure used for Hall-effect measurements.

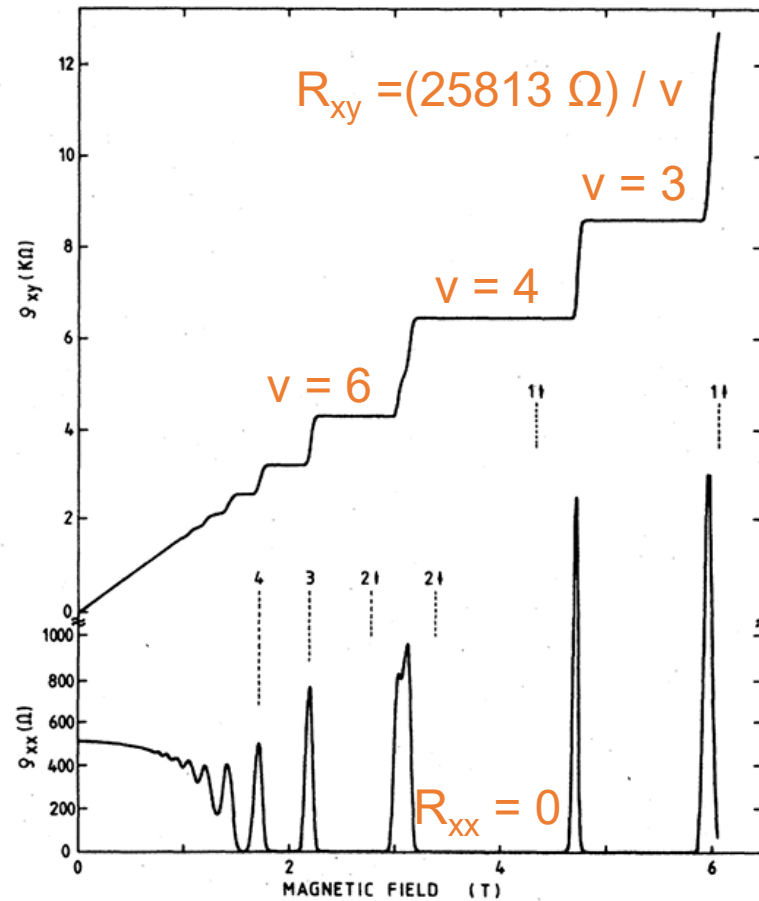


FIG. 14. Experimental curves for the Hall resistance $R_H = \rho_{xy}$ and the resistivity $\rho_{xx} \sim R_x$ of a heterostructure as a function of the magnetic field at a fixed carrier density corresponding to a gate voltage $V_g = 0$ V. The temperature is about 8 mK.

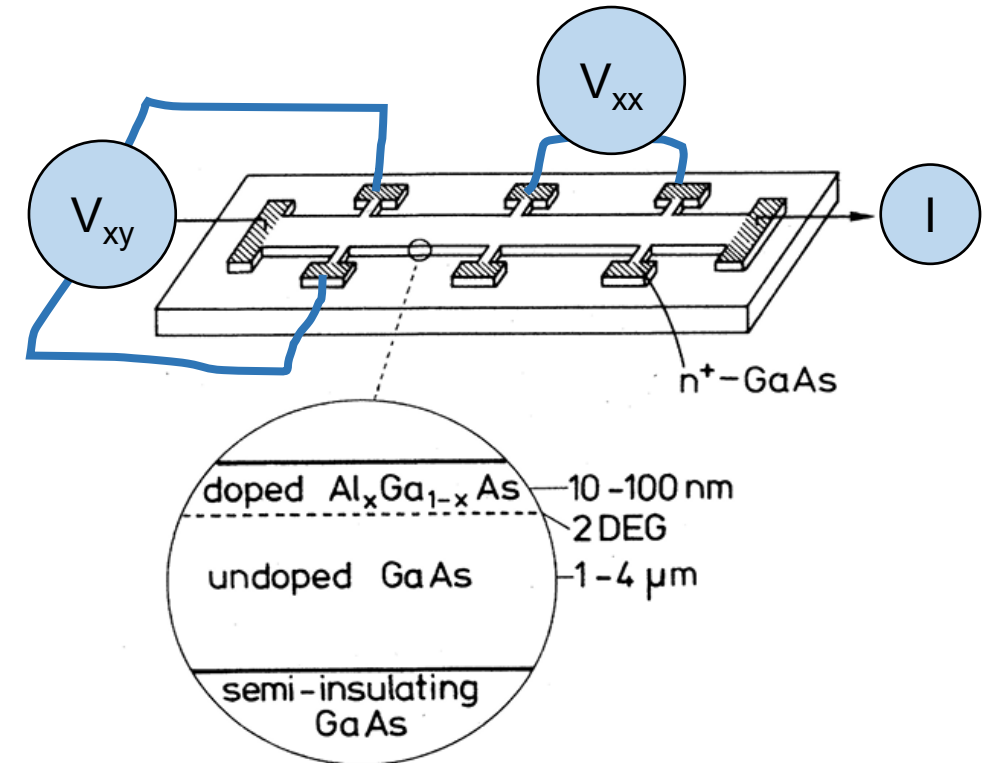


FIG. 3. Typical shape and cross section of a GaAs- $\text{Al}_x\text{Ga}_{1-x}\text{As}$ heterostructure used for Hall-effect measurements.

Zustände pro Landau Level (und Fläche)

$$n_{LL} = \frac{eB}{h}$$

Für feste Elektronendichte n_{2D} sind genau ν Landau Level gefüllt wenn gilt

$$n_{2D} = \frac{eB}{h} \cdot \nu$$

Mit dem klassischen Hallwiderstand erhalten wir

$$R_{xy} = R_H \cdot B = \frac{1}{e \cdot n_{2D}} \cdot B = \frac{h}{e^2} \cdot \frac{1}{\nu}$$



Experiment



$$n_{LL} = \frac{eB}{h}$$

Ohmsches Gesetz

$$\mathbf{j} = \sigma \cdot \mathbf{E} \quad \longrightarrow \quad \begin{pmatrix} j_x \\ j_y \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} \\ \sigma_{yx} & \sigma_{yy} \end{pmatrix} \cdot \begin{pmatrix} E_x \\ E_y \end{pmatrix}$$

Inversion des Leitfähigkeitstensors

$$\rho = \sigma^{-1}$$

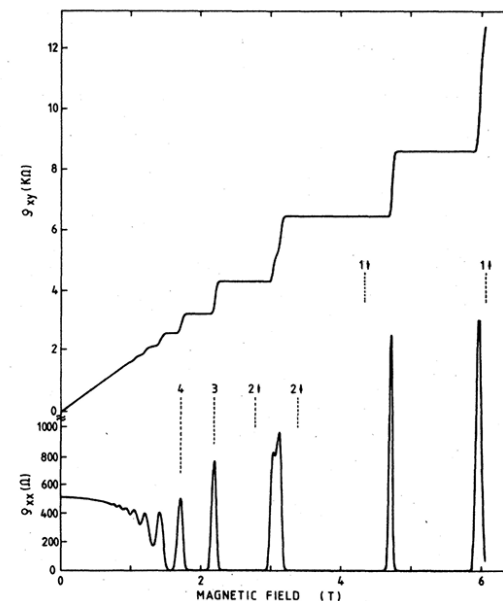
$$\sigma_{xx} = \frac{\rho_{xx}}{\rho_{xx}^2 + \rho_{xy}^2}$$

$$\rho_{xx} = \frac{\sigma_{xx}}{\sigma_{xx}^2 + \sigma_{xy}^2}$$

$$\sigma_{xy} = \frac{-\rho_{xy}}{\rho_{xx}^2 + \rho_{xy}^2}$$

$$\rho_{xy} = \frac{-\sigma_{xy}}{\sigma_{xx}^2 + \sigma_{xy}^2}$$

Im Experiment

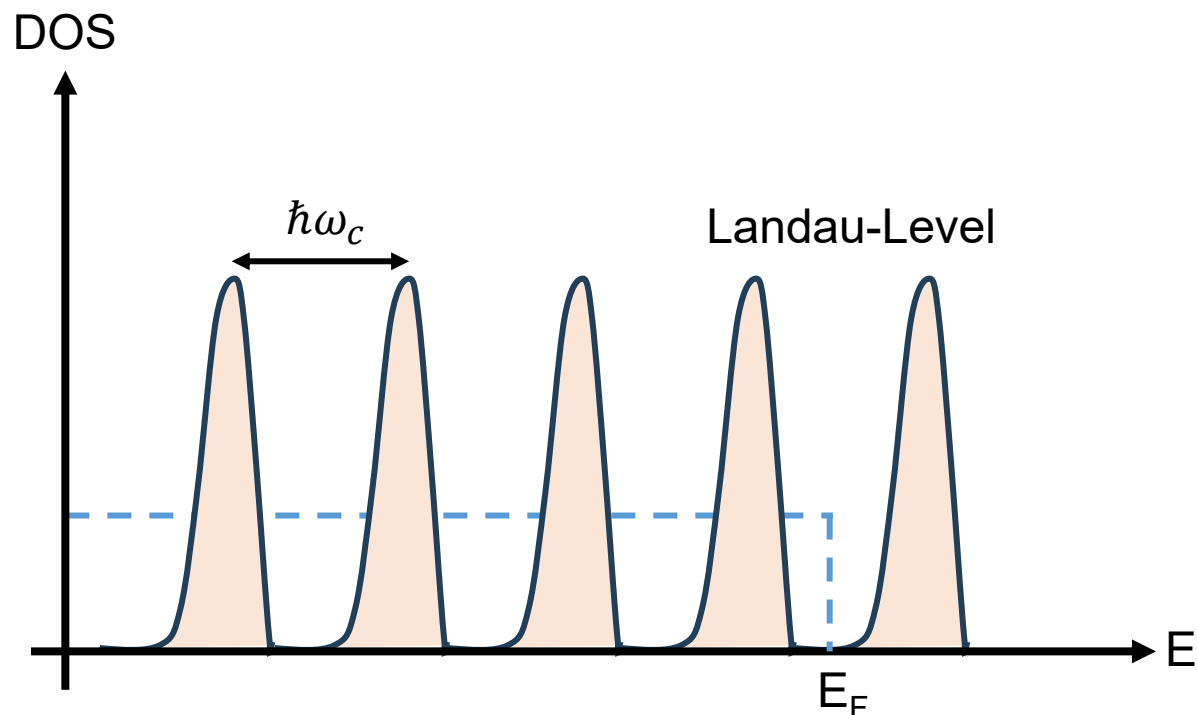


$$\sigma_{xy} = \frac{\frac{4e^2}{h}}{\left(\frac{4e^2}{h}\right)^2 + 0^2}$$

$$\sigma_{xx} = \frac{0}{\left(\frac{4e^2}{h}\right)^2 + 0^2} = 0$$



Landau-Level



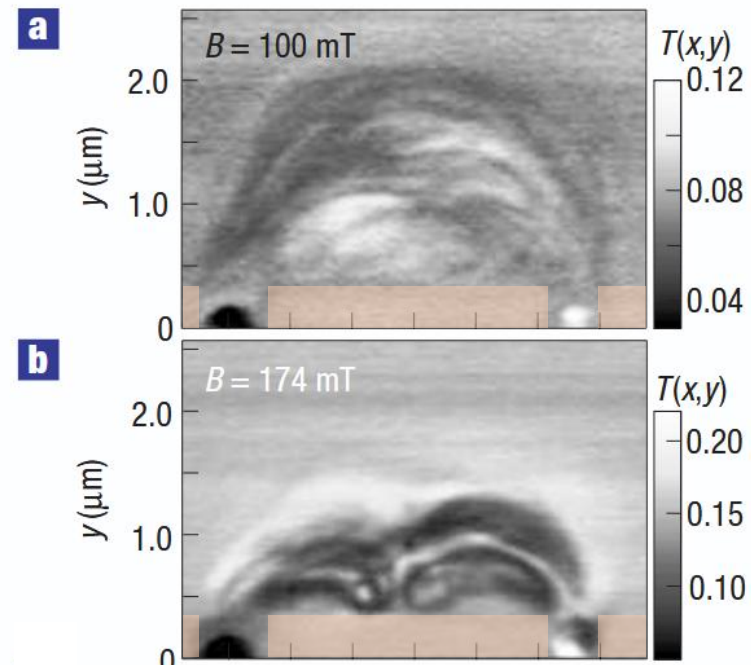
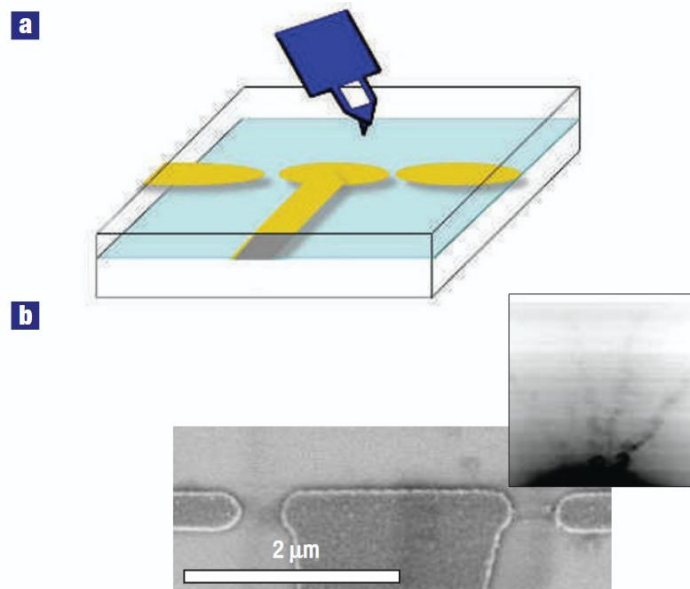
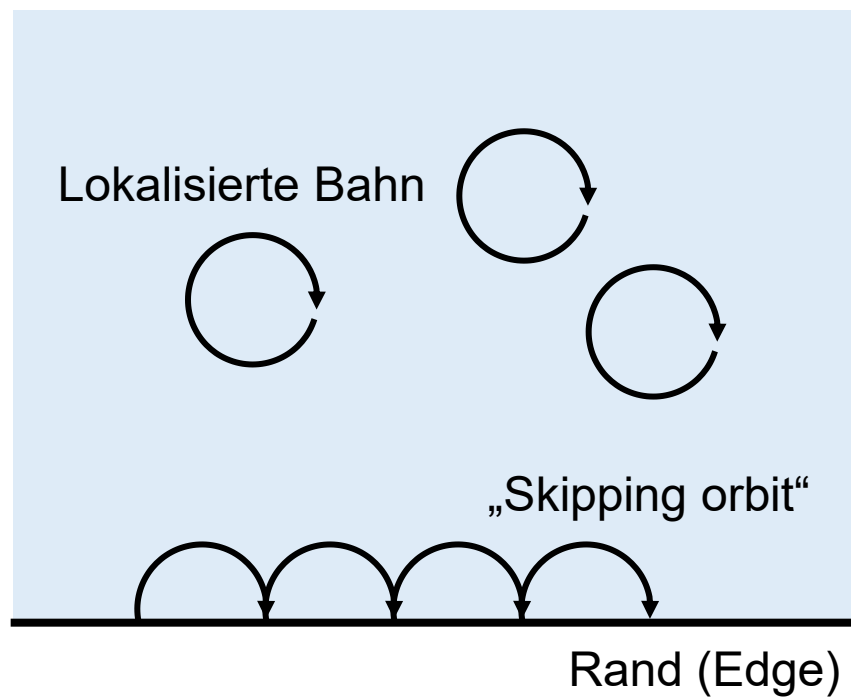
Konsistent bisher:

- Wert der Quantisierung
→ Landau-Level Entartung
- Fermi-Level zwischen Landau-Leveln
→ $\text{DOS}(E_F) \sim 0$ und $\sigma_{xx} = 0$

Wie fließt der Messtrom ohne eine Zustandsdichte bei E_F ?

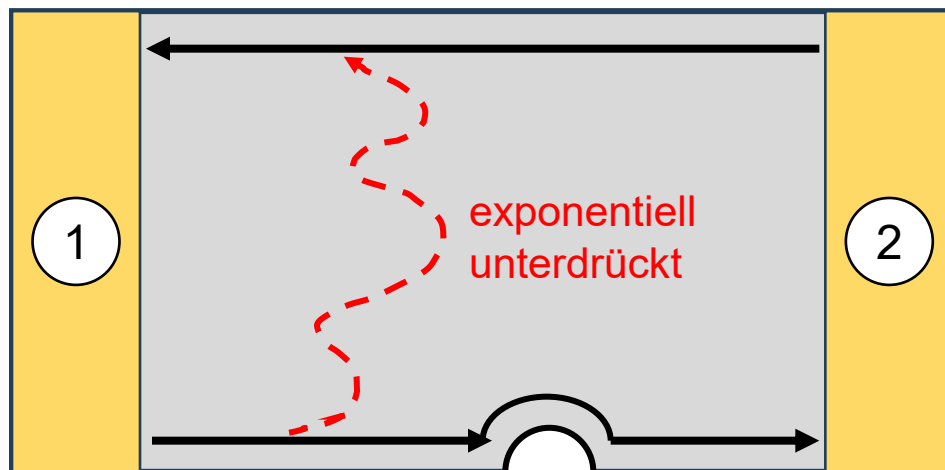
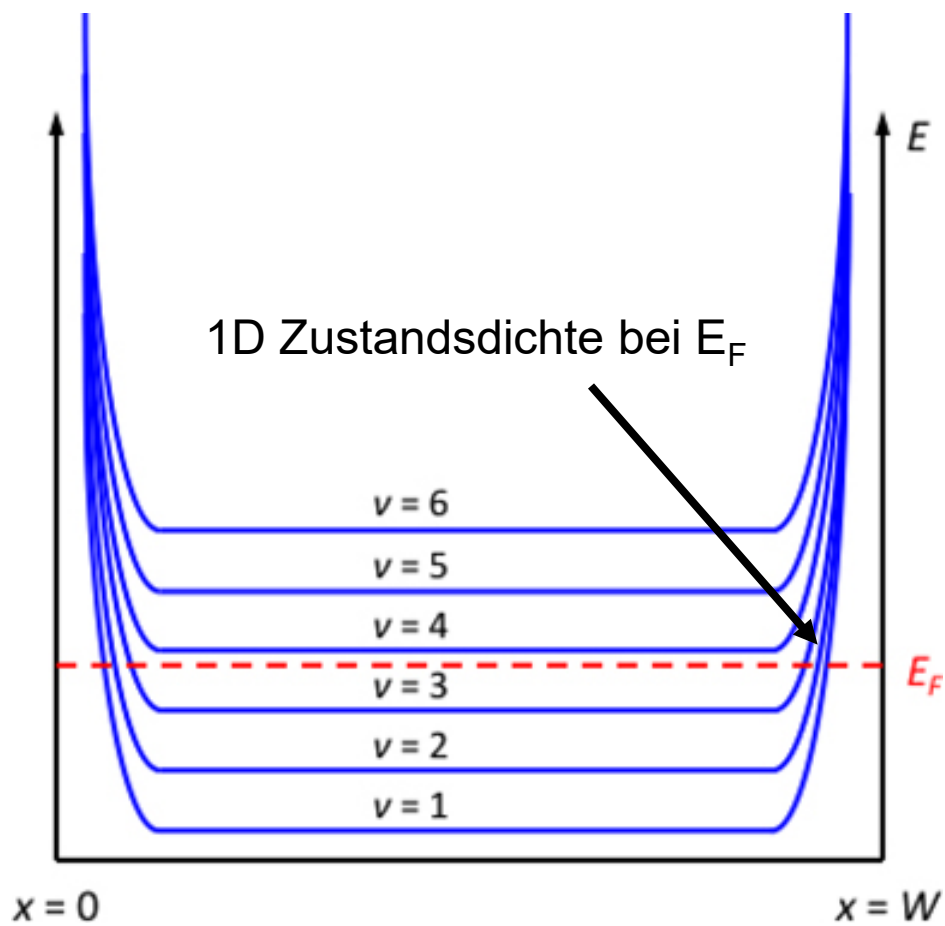
Wieso kann ein "Isolator" einen endlichen Hallwiderstand haben?

2D Volumen (Bulk)



Transport durch 1D Randkanäle

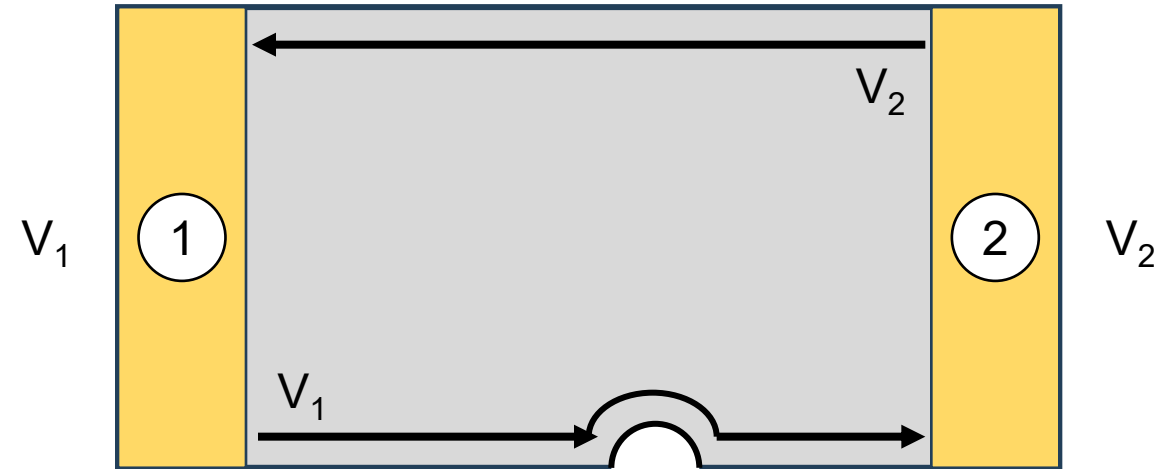
Rand durch Potentialwall angenähert



- 1D Randkanäle mit chiraler Propagation
→ keine Rückstreuung im Kanal
- Keine Streuung zwischen den Randzuständen durch die Volumenzustände
- Perfekte Transmission: $T_{12} = 1$ → Transport beschrieben durch *Landauer-Büttiker-Formalismus*

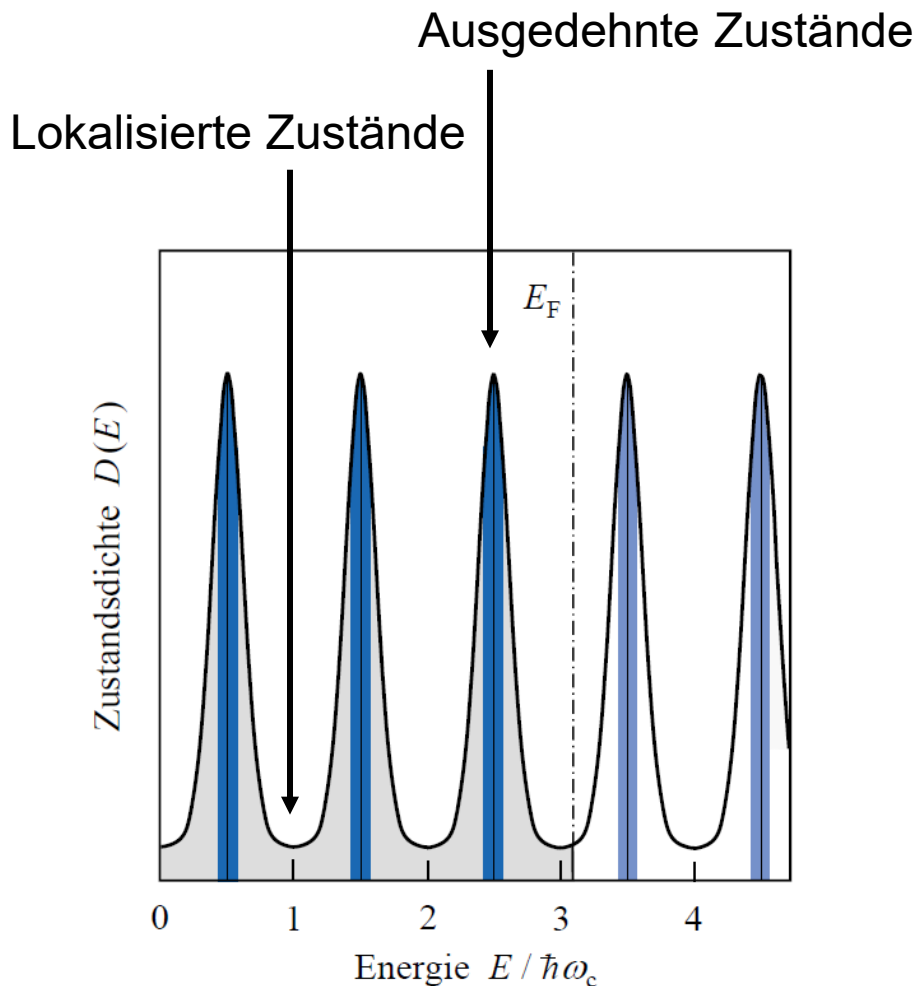
Der Strom ist durch die Anzahl der Randkanäle ν , das Leitswertquantum $\frac{e^2}{h}$ und die chemische Potentialdifferenz $V_1 - V_2$ gegeben

$$I_{12} = \nu \cdot T_{12} \cdot G_{1D} \cdot V_{12} = \nu \frac{e^2}{h} (V_1 - V_2)$$



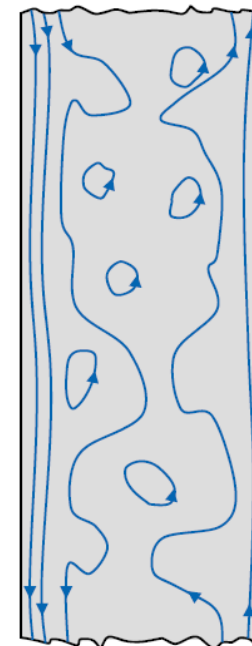
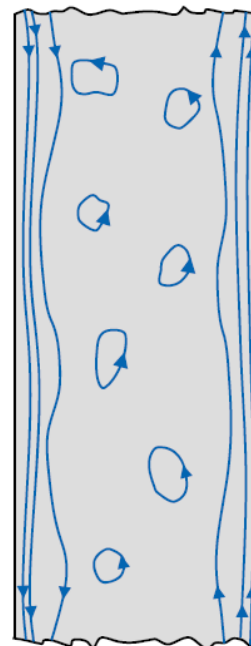
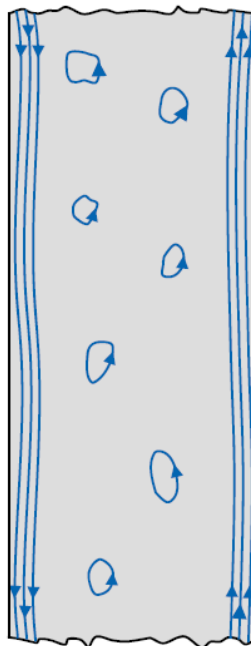
Da zwischen den Randkanälen keine Streuung stattfindet sind die Potentiale des „oberen“ und „unteren“ Randes durch V_2 und V_1 gegeben

$$R_{xy} = \frac{V_{xy}}{I} = \frac{V_1 - V_2}{\nu \frac{e^2}{h} (V_1 - V_2)} = \frac{1}{\nu} \cdot \frac{h}{e^2}$$



ρ_{xy} = quantisiert
 $\rho_{xx} = 0$

ρ_{xy} nicht quantisiert
 $\rho_{xx} \neq 0$



(a)

(b)

(c)

(d)

Bild 9.44: Elektronenzustände bei der Fermi-Energie. **a)** Neben den drei Randkanälen sind in der Mitte lokalisierte Zustände zu sehen. **b)** Mit zunehmendem Magnetfeld vergrößern sich die Bereiche mit lokalisierten Zuständen. Der innerste Kanal entfernt sich vom Probenrand. **c)** Inelastische Streuung zwischen den Elektronen der beiden innersten Kanäle tritt auf. **d)** Fermi-Energie und Landau-Niveaus fallen zusammen, der innerste Kanal ist in lokalisierte Zustände aufgebrochen. Der Längswiderstand ist maximal.

Nobel Prize in Physics 1998



Photo from the Nobel Foundation archive.

Robert B. Laughlin

Prize share: 1/3



Photo from the Nobel Foundation archive.

Horst L. Störmer

Prize share: 1/3



Photo from the Nobel Foundation archive.

Daniel C. Tsui

Prize share: 1/3

The Nobel Prize in Physics 1998 was awarded jointly to Robert B. Laughlin, Horst L. Störmer and Daniel C. Tsui "for their discovery of a new form of quantum fluid with fractionally charged excitations"

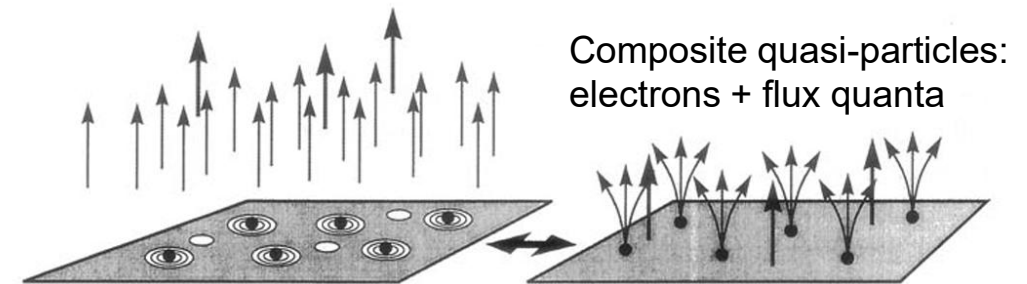
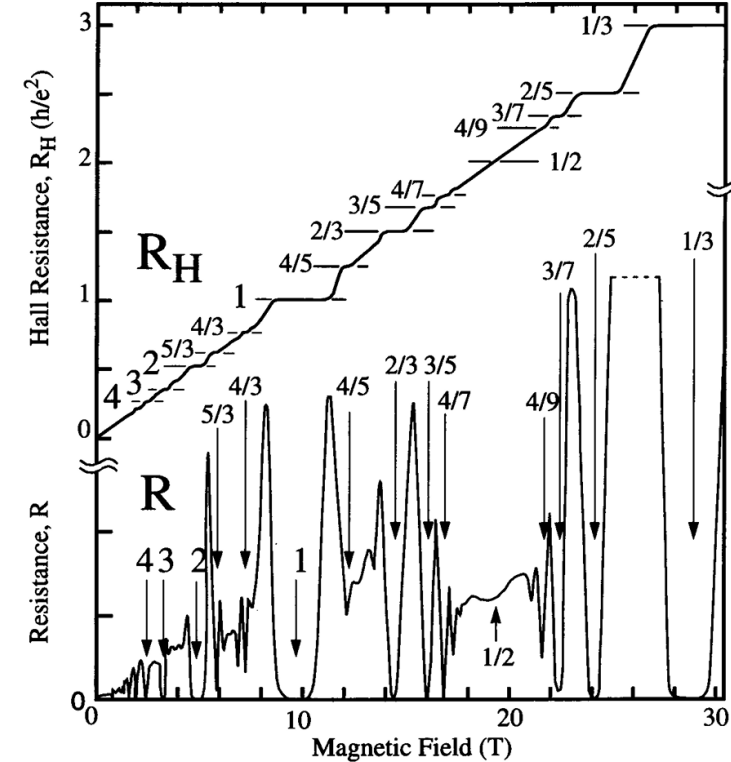


Figure 16. Schematic representation of $1/3$ charged quasiparticles. At slightly higher B fields than at $\nu=1/3$ additional vortices are created. They represent dimples in the electron lake. In the dimples exactly $1/3$ of an electron charge is missing. These are the fractionally charged quasi-particles of the FQHE.

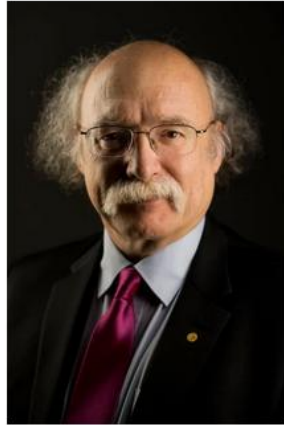
Nobel Prize in Physics 2016



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A. Mahmoud

David J. Thouless

Prize share: 1/2



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F. Duncan M.
Haldane

Prize share: 1/4



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A. Mahmoud

J. Michael Kosterlitz

Prize share: 1/4

The Nobel Prize in Physics 2016 was awarded with one half to David J. Thouless, and the other half to F. Duncan M. Haldane and J. Michael Kosterlitz "for theoretical discoveries of topological phase transitions and topological phases of matter"

$$\sigma_{xy} = C \cdot \frac{e^2}{h} = \sum_{n \text{ occ}} \int_{BZ} d^2k / (2\pi)^2 \Omega_n(\mathbf{k})$$

$$\Omega_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times A_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times i \langle u_{n\mathbf{k}} | \nabla_{\mathbf{k}} | u_{n\mathbf{k}} \rangle$$

- Chern number C
- Berry connection $A_n(\mathbf{k})$
- Berry curvature $\Omega_n(\mathbf{k})$

→ Hall conductance can be expressed in general geometric properties of the Hamiltonian

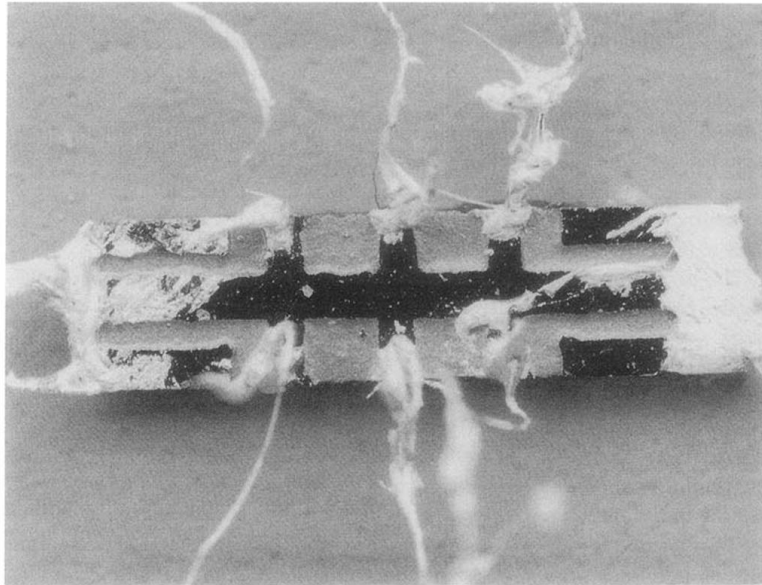
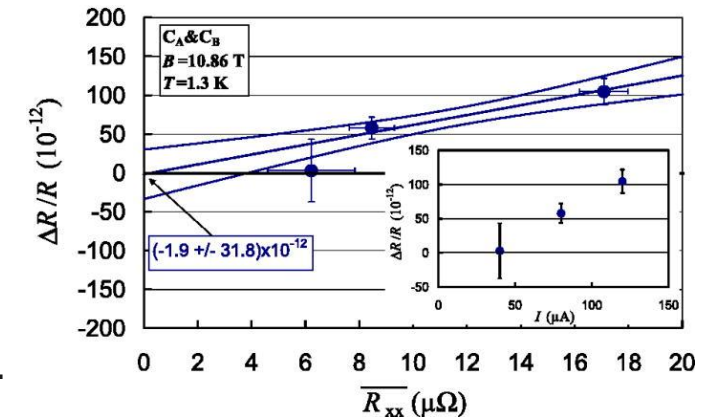
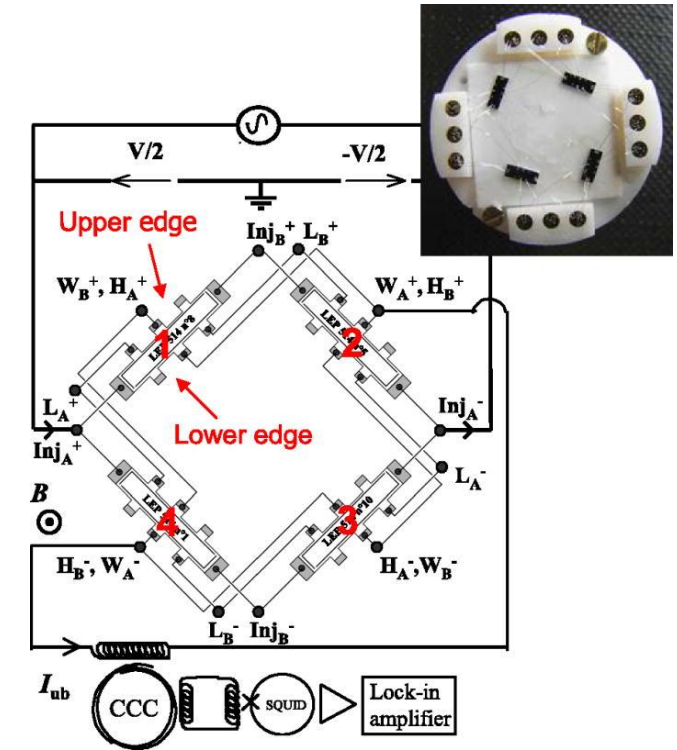
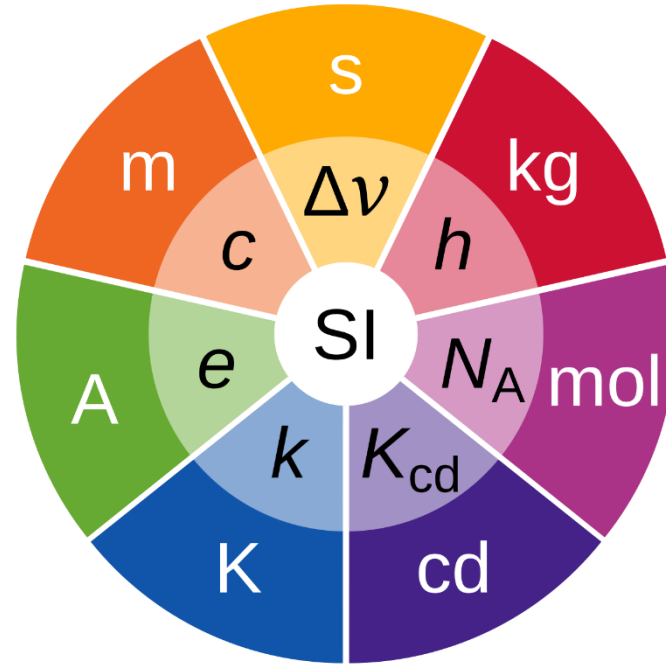
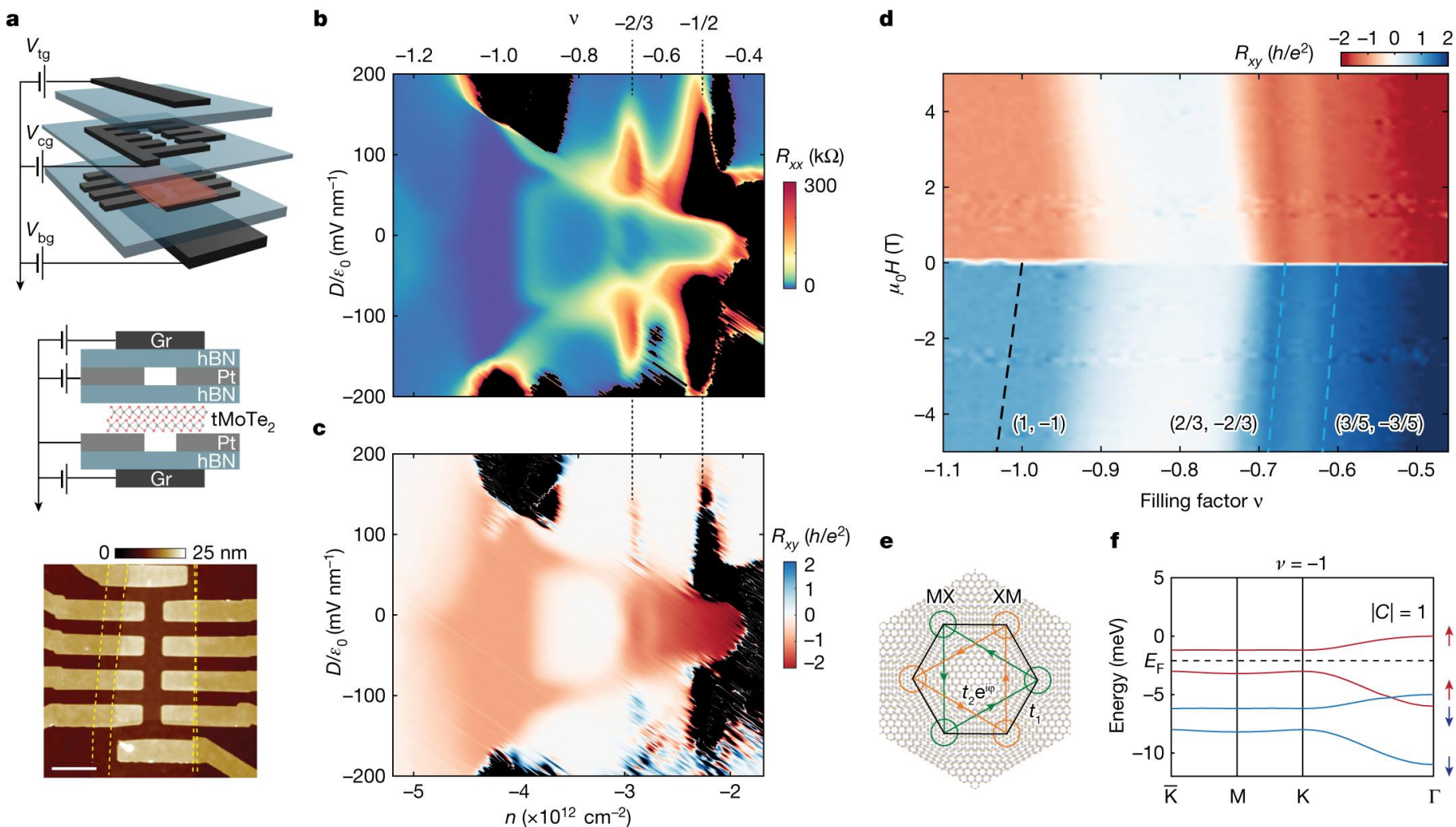


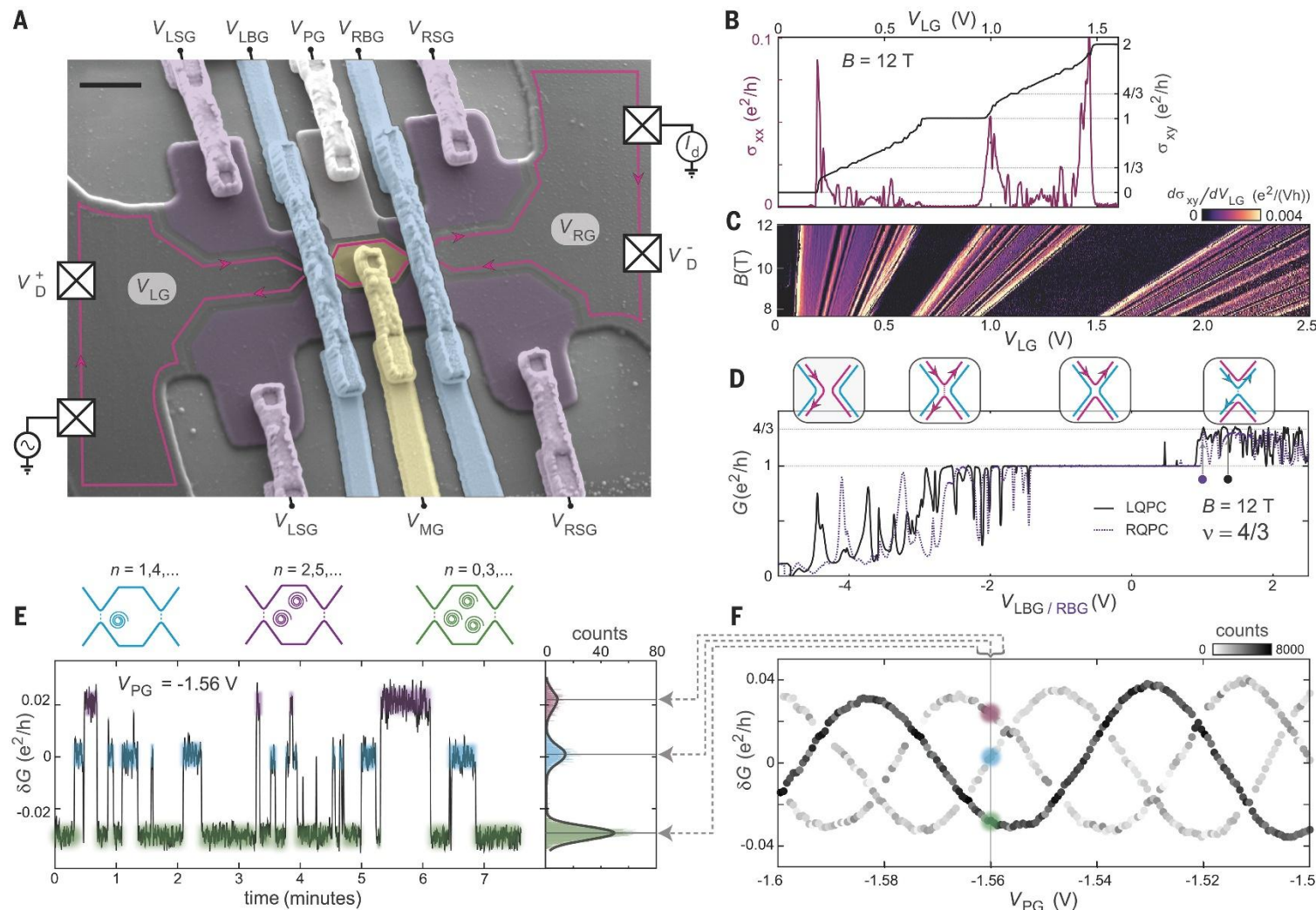
Figure 8. Photograph of a GaAs/AlGaAs sample. The size is about 6×1.5 mm. Black area (in reality mirror-like but reflecting the black camera) is the original surface above the 2DES. Gray areas have been scratched away to confine the current path to the center of the sample. White areas are indium blotches used to make contact to the 2DES. Gold wires are attached. Specimens like this one, prepared with little attention to exact dimension nor to tidiness, show quantization of the Hall resistance to an accuracy of a few 10 parts in a billion. The specimen shown is the sample in which the fractional quantum Hall effect (FQHE) was discovered in 1981.



Zum Beispiel: J. Appl. Phys. 114, 064508 (2013).



Nature volume 622, pages 74–79 (2023).



Anyonische Zustände bei geradzahlig, fraktionalem Füllfaktor, z.B. $5/2$, sind mögliche Kandidaten zur Realisierung von Topologischen Qubits.